Potential efficacy of broccoli sprouts as a unique supplement for management of type 2 diabetes and its complications.

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Functional foods and their nutraceutical components are now considered as supplementary treatments in type 2 diabetes and prevention of its long-term complications. Young broccoli sprouts as a functional food contain many bioactive compounds specially sulforaphane. In hyperglycemic and oxidative conditions, sulforaphane has the potential to activate the NF-E2-related factor-2 (Nrf2)-dependent antioxidant response-signaling pathway, induces phase 2 enzymes, attenuates oxidative stress, and inactivates nuclear factor kappa-B (NF-κB), a key modulator of inflammatory pathways. Interestingly, sulforaphane induces some peroxisome proliferator-activated receptors, which contribute to lipid metabolism and glucose homeostasis. In animal and in vitro models, sulforaphane also shows antihypertensive, anticancer, cardioprotective, and hypocholesterolemic capacity, and has bactericidal properties against Helicobacter pylori. Supplementation of type 2 diabetics with high sulforaphane content broccoli sprouts resulted in increased total antioxidant capacity of plasma and in decreased oxidative stress index, lipid peroxidation, serum triglycerides, oxidized low-density lipoprotein (LDL)/LDL-cholesterol ratio, serum insulin, insulin resistance, and serum high-sensitive C-reactive protein. Sulforaphane could prevent nephropathy, diabetes-induced fibrosis, and vascular complications. Potential efficacy of sulforaphane and probably other bioactive components of young broccoli sprouts makes it as an excellent choice for supplementary treatment in type 2 diabetes.